

QuantumSafe Analyzer

Cryptographic Risk Assessment Report

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Executive Summary

This assessment analyzed the organization's cryptographic inventory to identify systems vulnerable to quantum computing attacks. **Total Assets:** 15

High Risk Assets: 11

Low Risk Assets: 4

Average Risk Score: 7.67

Harvest Now, Decrypt Later Risks: 9

Risk Assessment Results

Asset	Algorithm	Risk	Score	Recommendation
Internet Banking	RSA	High	10	Replace with ML-KEM Hybrid
Mobile Banking	ECC	High	9	Replace with ML-KEM + ML-DSA
ATM Network	RSA	High	10	Replace with ML-KEM Hybrid
Payment Gateway	RSA	High	10	Replace with ML-KEM Hybrid
Employee Email	RSA	High	10	Replace with ML-KEM Hybrid
VPN Gateway	RSA	High	10	Replace with ML-KEM Hybrid
Customer Database	AES	Low	2	Continue using AES-256
Backup Server	AES	Low	2	Continue using AES-256
Cloud Storage	AES	Low	2	Continue using AES-256
Certificate Authority	RSA	High	10	Replace with ML-KEM Hybrid
Internal Portal	ECC	High	9	Replace with ML-KEM + ML-DSA
Document Management	RSA	High	10	Replace with ML-KEM Hybrid
Core Banking System	RSA	High	10	Replace with ML-KEM Hybrid
Fraud Detection Server	AES	Low	2	Continue using AES-256
Identity Management	ECC	High	9	Replace with ML-KEM + ML-DSA

Recommendations

- Replace RSA with ML-KEM Hybrid Deployment.
- Replace ECC Digital Signatures with ML-DSA.
- Continue using AES-256 for symmetric encryption.

- Deploy Hybrid TLS during migration.
- Prioritize long-term confidential systems for migration.
- Improve key lifecycle management using HSMs and Key Vaults.

Conclusion

The organization currently relies on several cryptographic systems that are vulnerable to future quantum attacks. A phased migration towards Post-Quantum Cryptography (PQC) using hybrid cryptographic deployment is recommended to maintain long-term confidentiality and regulatory compliance.